



filip.morys@mcgill.ca



NW252, 3801 Rue University, H3A 2B4, Montreal, Quebec, Canada



<https://filipmorys.github.io/>

EDUCATION

- 2013-2019** **PhD Student/Researcher**
Integrated Research and Treatment Center Adiposity Diseases, Leipzig
University Medical Center;
Max Planck Institute for Human Cognitive and Brain Sciences, Decision-
making in obesity: neurobiology, behavior and plasticity group (Dr. Annette
Horstmann), Leipzig, Germany
PhD thesis: Characterising and altering maladaptive behaviours and
tendencies in obesity
- 2011-2013** **Master's Degree**
International Graduate Program Medical Neurosciences
Charité – Medical University, Berlin, Germany
Master's thesis: Decoding task sets from human brain activity
- 2008-2011** **Bachelor's Degree**
Neurobiology
Jagiellonian University, Cracow, Poland
Bachelor's thesis: Neuropsychology of prospective memory

PROFESSIONAL EXPERIENCE

- 2019-currently** **Postdoctoral Fellow/Research Associate**
Montreal Neurological Institute, McGill University, Montreal, Canada
- 2018-2020** **Research Scientist**
CodersFirst, Warsaw, Poland.
Development of cognitive and personality testing for IT recruitment
- 2012-2013** **Research Master's Thesis**
Bernstein Center for Computational Neuroscience, Laboratory of Theory and
Analysis of Large-Scale Brain Signals (Prof. Dr. John-Dylan Haynes); Berlin,
Germany
- 2012** **Laboratory Rotation**
Dahlem Institute for Neuroimaging of Emotions (Prof. Dr. Felix Blankenburg);
Berlin, Germany
- 2012** **Laboratory Rotation**
Charité – Medical University Berlin, Working Group Cognition, Ageing,
Neurorehabilitation (Prof. Dr. Agnes Flöel); Berlin, Germany
- 2010** **Internship**
Nencki Institute for Experimental Biology, Laboratory of Preclinical Studies for
Neurodegenerative Diseases (Prof. Dr. Grazyna Niewiadomska); Warsaw,
Poland

FUNDING

2025	Building a Montreal Hub for Neurometabolic Discovery. Budget: CAD 30,000 PI: Paul Sabatini Co-Applicants: Dana Small, Stephanie Fulton, Filip Morys The Research Network on cardiometabolic health, diabetes and obesity: Strategic Workshop 2025 competition
2025	Associations between neurocognitive health and weight loss in children and adults undergoing bariatric surgery Budget: CAD 199,765 PI: Dana Small, Filip Morys Co-applicants: Julius Erdstein, Michael Tsoukas, David Labbé, Chelsia Gillis, Irah King, Xiao Wen Hu, Cristian O'Flaherty, Isabelle J. Gagnon, Vincent Mooser, A. Dagher, Angela Alberga, Etienne St-Louis, Augusto Montezano, Vanessa Tardio, Rhian Touyz, Amin Andalib, Sridar Narayanan, Diego Herrera, Amanpreet Badhwar, AmanPreet Badhwar, Rachel Massicotte, Stephanie Chevalier, May Faraj, Ian Patton, Indra Gupta, Kaberi Dasgupta, Patricia Tak Sam Li
2025	Effects of GLP1 agonist-induced weight loss on cerebral and blood markers of neurodegeneration in individuals with obesity and genetic risk for Alzheimer's disease. Budget: CAD 1,057,080 PI: Alain Dagher Co-Applicants: Filip Morys, Dana Small, Claudine Gauthier, Christine Tardif, Yashar Zeighami, Xiao Wen Hu, Sylvia Villeneuve, Natasha Rajah, Ian Patton Canadian Institutes of Health Research, Team Grant, Canadian Consortium on Neurodegeneration in Aging (CCNA) Phase III: Research Teams
2025	Effects of GLP1-agonist induced weight loss on brain, metabolic, and cerebrovascular health measures. Budget: CAD 1,346,400 PI: Alain Dagher Co-Applicants: Filip Morys, Dana Small, Stephanie Fulton, Claudine Gauthier, Christine Tardif, Yashar Zeighami, Xiao Wen Hu, Bratislav Mistic, Michael Tsoukas Canadian Institutes of Health Research. Project Grant Fall 2024.
2024	An MRI study of metabolic and cerebrovascular health Budget: CAD 10,000 PI: Filip Morys, Alain Dagher Canada Excellence Research Chair Symposium Pilot Project Award
2023	Measuring the effects of poor metabolic health on the brain Budget: CAD 50,000 PI: Alain Dagher Co-Applicants: Filip Morys, Bratislav Mistic, Claudine Gauthier, Christine Tardif, Michael Tsoukas, Xiao Wen Hu The Cognitive Neuroscience Knowledge Exchange for Clinical Translation (CONNECT) project. Healthy Brains for Healthy Lives.

2023	Advanced MRI measures of obesity-related brain damage. Budget: CAD 25,000 PI: Alain Dagher Co-Applicants: Filip Morys, Claudine Gauthier, Christine Tardif McConnell Brain Imaging Centre Collaboration Program FY24
2022-2023	Fonds de Recherche Quebec – Santé Postdoctoral Fellowship Budget: CAD 90,000 Main applicant: Filip Morys Supervisor: Alain Dagher

KEY PUBLICATIONS

1. **Morys F***, Kanyamibwa A*, Fängström D, Tweedale M, Pastor-Bernier A, Azizi H, Liu L, Horstmann A, Dagher A. Ultra-processed food consumption affects structural integrity of feeding-related brain regions independent and via adiposity. *Npj Metabolic Health and Disease*. 2025.
2. **Morys F**, Liu L, Senkevich K, Gan-Or Z, Dagher A. Obesity-related brain atrophy is independent of Alzheimer's disease protein pathways. *Journal of Alzheimer's Disease*. 2025.
3. **Morys F**, Tremblay C, Rahayel S, Hansen JY, Dai A, Misic B, et al. Neural correlates of obesity across the lifespan. *Communications Biology*. 2024.
4. **Morys F**, Potvin O, Zeighami Y, Vogel J, Lamontagne-Caron R, Duchesne S, Dagher A. Obesity-Associated Neurodegeneration Pattern Mimics Alzheimer's Disease in an Observational Cohort Study. *Journal of Alzheimer's Disease*. 2023.
5. **Morys F**, Dadar M, Dagher A. Association between mid-life obesity, its metabolic consequences, cerebrovascular disease and cognitive decline. *Journal of Clinical Endocrinology and Metabolism*. 2021.

BOOK CHAPTERS

1. Dagher A., **Morys F.**, Garcia-Garcia I. Body Brain Interactions in the Control of Feeding. In JH. Grafman (Ed.) *Encyclopedia of the Human Brain* (2nd ed., pp. 440-451). Elsevier. 2024.

LEADERSHIP

2026	Co-chair – Montreal Neurometabolic Seminar Series
2026	Founder/co-chair – Participant Advisory Committee; Canada Excellence Research Chair in Metabolism and the Brain
2025-2028	Program Committee – Society for the Study of Ingestive Behavior
2025-currently	Chair – Diversity, Equity, and Inclusivity in Action Committee; Canada Excellence Research Chair in Metabolism and the Brain
2024	Mentor – Maximize your Research on Obesity and Diabetes platform launch event. 2024, Toronto, Canada
2022-2024	Chair – multiple symposia during the Society for the Study of Ingestive Behavior Annual Meetings
2022	Organizer – Re-envisioning the future of academic training. Co-organizers: Carolina Makowski, Phoebe Thomson, Anne Bilot. Organization for Human Brain Mapping Annual Meeting 2022, Glasgow, United Kingdom.

2022	Chair/Organizer – Becoming a Leading Investigator in Human Neuroscience. Symposium organized by the Student and Postdoc Special Interest Group of the Organization for Human Brain Mapping during the Annual Meeting 2022, Glasgow, United Kingdom.
2022	Chair/Organizer – Lunch with mentors. In-person mentoring event organized by the Student and Postdoc Special Interest Group of the Organization for Human Brain Mapping during the Annual Meeting 2022, Glasgow, United Kingdom.
2021	Chair/Organizer - Link with mentors. Online mentoring event organized by the Student and Postdoc Special Interest Group of the Organization for Human Brain Mapping.

ACADEMIC SERVICE & OUTREACH

2026	Data Quartet Challenge – manuscript reviewer
2026	Volunteer organizer of Sadie's Walk – a community type 2 diabetes awareness event in Kahnawake Indigenous community, Québec
2023; 2026	Associate Editor – Journal of Alzheimer's Disease
2025-currently	Mentorship committee member of MyRoad - Maximize your Research on Obesity and Diabetes
2024	Guest Editor – Frontiers in Neuroscience
2020-2022	Secretary of the Student and Postdoc Special Interest Group of the Organization for Human Brain Mapping
2020-2022	Member of the Postgraduate Student Society Equity and Diversity Committee
2020-currently	Organization for Human Brain Mapping meeting abstract reviewer
2021	Organization for Human Brain Mapping Annual Meeting organizer and moderator of workshops and symposia of the Student and Postdoc Special Interest Group
2020	Neuromatch 2.0 Conference organization and abstract reviewer
2016-2018	Member of the Data Quality Assurance Team in the Neurology Department at the Max Planck Institute for Human Cognitive and Brain Sciences; Leipzig, Germany
2017	Organisation of International Research Training Group Autumn School, Leipzig, Germany
2012-2015	Writer and editor of the Charité Neuroscience Newsletter, a quarterly magazine on the newest topics in neuroscience
2012-2013	Creator of a handbook of Medical Neurosciences for PhD students
2011	Organization of the International Brain Research Organization Neuroscience Forum Neuronus, Jagiellonian University; Cracow, Poland

AWARDS & HONOURS

2023	Society for the Study of Ingestive Behavior New Investigator Travel Award for the project: 'Influence of obesity and MAPT/ApoE genotypes on brain structure and cognition' – 500 USD
2022	Relève étoile Jacques Genest award for best publication – 1,000 CAD
2022	Lewis Reford Travel Award, Montreal Neurological Institute – 1,500 CAD
2021	Harry R. Kissileff award for the best short oral presentation at the Annual Meeting of the Society for the Study of Ingestive Behavior
2020	Lewis Reford Travel Award, Montreal Neurological Institute – 1,500 CAD

2018	Research Academy Leipzig Travel Grant – 715 EUR
2017	Grant: Understanding the contribution of hemispheric asymmetry to overeating behaviour in obesity. PIs: Dr. Lieneke Janssen, Dr. Annette Horstmann. German Federal Ministry of Education and Research. Role: Conceptualisation of the proposed project.
2017	Research Academy Leipzig Travel Grant – 135 EUR
2017	Society for the Study of Ingestive Behavior New Investigator Travel Award for the project: ‘Neural correlates of retraining automatic action tendencies in obesity’ – 1,000 USD
2017	German Obesity Society best poster award for the project: ‘Neural correlates of retraining automatic action tendencies in obesity’ – 500 EUR
2014	Travel grant and stipend to attend Obesity Summer Bootcamp 2014 – Leipzig University – 3,000 EUR
2012/2013	Fellowship for up-and-coming scientists by the NeuroCure Cluster of Excellence; Berlin, Germany – 11,000 EUR

TEACHING

2025	Basics of fMRI and fMRI analysis. Montreal Neurological Institute, McGill University, Montreal, Canada. 2025-10 – 2025-11.
2018	fMRI analysis with SPM. Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Germany, 2018-09 – 2018-10.
2015	Maladaptive plasticity in behavioral disorders and addictions. Lecture. Berlin School of Mind and Brain, Humboldt University Berlin, Germany, 2015-10-06 – 2015-10-08.

INVITED TALKS

2026	<u>Upcoming</u> : <i>Round Table: Why Inclusivity? The imperative to change</i> . Talk to be presented at the Mentorship for Inclusive Science Summer School . Montreal, Canada.
2026	<u>Upcoming</u> : <i>Brain-obesity interactions across the lifespan: insights from population-based cohorts</i> . Talk to be presented at the 2026 Neurometabolic Summer Seminars by the SciN. Montreal, Canada.
2026	<i>Obesity, Metabolic Dysfunction, Environment, and the Brain: Evidence from Population-Based Cohorts Across the Lifespan</i> . Talk presented at the 11th Canadian Neurometabolic Meeting . Montreal, Canada.
2026	<i>Neurobehavioural causes and consequences of obesity in population-based cohorts</i> . Society of Biological Psychiatry Annual Meeting Symposium , New York City, USA. (https://doi.org/10.1016/j.biopsych.2026.03.063)
2025	<i>Obesity, metabolism, diet, and brain health</i> . Neurology Grand Rounds. The Montreal Neurological Institute , McGill University, Montreal, Canada.
2025	<i>Ultra-processed food consumption affects structural integrity of feeding-related brain regions independent and via adiposity</i> . Modern Diet and Physiology Research Center , McGill University, Montreal, Canada.

2024	<i>Public lecture – Reshaping the brain: exploring the weight loss effects of modern medication.</i> Research Institute of the McGill University Health Center. Montreal, Canada.
2024	<i>Brain-obesity interactions across the lifespan: insights from population-based cohorts.</i> New York University. New York, USA.
2024	<i>Neurocognitive correlates of chronic obesity.</i> Society for the Study of Ingestive Behavior. Chicago, USA.
2024	<i>Methods of neuroscientific inquiry.</i> Online talk presented in a Polish high school for Brain Awareness Week. Tarnowskie Gory, Poland.
2023	<i>Mindful of your waistline: the relationship between obesity, brain atrophy, and cognitive decline.</i> MoCA Cognition. Online seminar for health practitioners.
2023	<i>Neural correlates of obesity across the lifespan.</i> Centre Hospitalier Universitaire Sainte-Justine. Montreal, Canada.
2021	<i>Navigating academic career after PhD.</i> Alumni talk series. Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Germany.
2020	<i>Influence of metabolic syndrome on brain and cognition.</i> Modern Diet and Physiology Research Center, Yale University, New Haven, USA.

SUPERVISION AND MENTORING

2025	Justin Goldberg (graduate, MSc program)
2024-currently	Shilan Heshmati (graduate, PhD program)
2023-2025	Mayuri Maheswaran (graduate, MSc program)
2023-2024	Karl Wennberg (research assistant)
2023-2024	Max Tweedale (undergraduate)
2022	Yuxin Zhou (undergraduate)
2022	Elena Ock (undergraduate)
2021-2022	Jessica Yu (graduate, MSc program)
2017	Christina Breil (undergraduate)
2014	Parniane Arbab-Zadeh (undergraduate)
2014	Maram Amara (undergraduate)
2014	Mailin Krüger (undergraduate)

REMAINING PUBLICATIONS

1. Kouchache T, Zolfaghari S, Bouhadoun S, Delva A, Pelletier A, Kuhlencord M, Noyce A, Waters S, Belete D, Wilkinson T, Bush K, **Morys F** et al. Reply to comment on “delayed disease onset report in UK biobank: Implications for prodromal studies in Parkinson’s disease.” *Movement Disorders*. 2026.
2. Azizi H, Abbasi N, Liu L, Pastor-Bernier A, Tremblay C, Pourmajidian M, Senkevich K, **Morys F**, Vo, Moqadam R, Rajimehr R, Yu E, Savadjiev P, Markello RD, Shafiei G, Khatibi N, Jahanshad N, Thompson PM, Poline JB, Gan-Or Z, Misic B, Zeighami Y, Dagher A. Brain size links neurodevelopment to neurodegeneration in Parkinson’s disease. *bioRxiv*. 2025
3. Farahani A, Liu Z-Q, **Morys F**, et al. Joint multi-omics profiling of brain and body health in aging. *bioRxiv*. 2025.
4. Zolfaghari S, Kouchache T, Delva A, Bouhadoun S, Kuhlencord M, Pelletier A, Noyce AJ, Waters S, Belete D, Wilkinson T, Bush K, **Morys F**, Vo A, Rannikmae A, Dagher A, Postuma RB. Delayed disease

- onset report in UK Biobank: Implications for prodromal studies in Parkinson's disease. *Movement Disorders*. 2025.
5. Quillian J, Attuquayefio T, Sung J, Canna A, Ko T, Davis X, Maciejewski K, Li F, Santoro N, Pierpont B, Kullmann S, Preissl H, **Morys F**, Dagher A, Caprio S, Small D. Neurocognition in youth with versus without prediabetes. *medRxiv*. 2025.
 6. Shishikura M, Liu L, Yu E, Zhu R, Vilar-Ribó L, Cupertino R, Palmer A, Sanchez-Roige S, Vainik U, **Morys F**, Gan-Or Z, Misic B, Dagher A. Genomic structural equation modeling of impulsivity and risk-taking traits reveals three latent factors distinctly associated with brain structure and development. *bioRxiv*. 2025.
 7. Merz E, **Morys F**, Hansen M, Strack J, Jacobs L, Vainik U, Shishikura M, Myers B. Socioeconomic factors, brain-derived neurotrophic factor Val66Met polymorphism, and cortical structure in children and adolescents. *Scientific Reports*. 2025.
 8. Adise S, Li ZA, Ottino-González J, **Morys F**, Chiarelli PA, Hershey T. Distinct patterns of weight gain, age, and subcortical microstructure in early adolescence. *JAMA Netw Open*. 2025.
 9. Tremblay C, Rahayel S, Pastor-Bernier A, St-Onge F, Vo A, Rheault F, Daneault V, **Morys F**, Rajah N, Villeneuve S, Dagher A. Pattern of mechanisms of atrophy progression in individuals with a family history of Alzheimer's disease: a comparative study. *Brain Communications*. 2025
 10. Bouhadoun S, Zolfaghari S, Delva A, Pelletier A, Kouchache T, Dagher A, **Morys F**, Vo A, Kuhlencord M, Postuma R. Exploring the Nicotine-Parkinson's disease link – Insights from the UK Biobank. *Parkinsonism and Related Disorders*. 2024.
 11. **Morys F***, Tweedale M*, Pastor-Bernier A, Azizi H, Tremblay C, Dagher A. Obesity and diffusion-weighted imaging of subcortical grey matter in young and older adults. *Appetite*. 2024.
 12. **Morys F**, Yu E, Shishikura M, Paquola C, Vainik U, Nave G, Koellinger P, Gan-Or Z, Dagher A. Neuroanatomical Correlates of Genetic Risk for Obesity in Children. *Translational Psychiatry*. 2023.
 13. Yu J, **Morys F**, Dagher A, Lajoie A, Gomes T, Ock E, Kimoff RJ, Kaminska M. Associations between sleep-related symptoms, obesity, cardiometabolic conditions, brain structural alterations and cognition in the UK biobank. *Sleep Medicine*. 2023.
 14. Hartmann H, Janssen L, Herzog N, **Morys F**, Fängström D, Fallon SJ, Horstmann A. Self-reported intake of high-fat and high-sugar diet is not associated with cognitive stability and flexibility in healthy men. *Appetite*. 2023.
 15. **Morys F***, Shishikura M*, Dagher A. Population-based research in obesity – an overview of neuroimaging studies using big data approach. *Current Opinion in Endocrine and Metabolic Research*. 2022.
 16. Yang Y, Qi L, **Morys F**, Wu Q, Chen H. Food-specific inhibition training for food devaluation: A meta-analysis. *Nutrients*. 2022
 17. García-García I, Michaud A, María-Jurado Á, Dagher A, **Morys F**. Mechanisms linking obesity and its metabolic comorbidities with cerebral grey and white matter changes. *Reviews in Endocrine and Metabolic Disorders*. 2022.
 18. **Morys F***, Poessel M*, Breuer N*, Villringer A, Hummel T, Horstmann A: Brain response to food odors is not associated with body mass index and obesity-related metabolic health measures. *Appetite*. 2022.
 19. Yang Y, **Morys F**, Wu Q, Li J, Chen H. Pilot study of food-specific go/no-go training for overweight individuals: brain imaging data suggest inhibition shapes food evaluation. *Social Cognitive and Affective Neuroscience*. 2021.
 20. Yang Y, Wu Q, **Morys F**. Brain Responses to High-Calorie Visual Food Cues in Individuals with Normal-Weight or Obesity: An Activation Likelihood Estimation Meta-Analysis. *Brain Sciences*. 2021.

21. Beyer F, Horn K, Frenzel S, Hofer E, Knol MJ, **Morys F**, Vainik U, Auwera S Van der, Wittfeld K, Saba Y, et al.: Investigating the genetic and environmental basis of head micromovements during MRI. *bioRxiv*. 2021.
22. **Morys F***, Garcia-Garcia I*, Neseliler S*, Dadar M, Yau Y, Scala S et al. Relationship between impulsivity, uncontrolled eating and body mass index: a hierarchical model. *International Journal of Obesity*. 2021.
23. Tremblay C, Rahayel S, Vo A, **Morys F**. et al. Brain atrophy progression in Parkinson's disease is shaped by connectivity and local vulnerability. *Brain Communications*. 2021.
24. **Morys F**, Dagher A. Metabolic syndrome increases COVID-19-related mortality in the UK Biobank sample. *Frontiers in Endocrinology*. 2021.
25. Park B, Park H, **Morys F**, Kim M, Byeon K, Lee H et al. Body mass variations relate to fractionated functional brain hierarchies. *Communications Biology*. 2020
26. **Morys F**, García-García I, Dagher A. Is obesity related to enhanced neural reactivity to visual food cues? A review and meta-analysis. *Social Cognitive and Affective Neuroscience* 2020.
27. García-García I, **Morys F**, Michaud A, Dagher A. Food addiction, skating on thin ice. A critical overview of neuroimaging findings. *Current Addiction Reports*. 2020.
28. Park B, Byeon K, Lee MJ, Chung CS, Kim SH, **Morys F** et al. Whole-brain functional connectivity correlates of obesity phenotypes. *Human Brain Mapping*. 2020.
29. **Morys F**, Simmank J, Horstmann A. DRD2/ANKK1 Taq1A but not COMT single nucleotide polymorphisms contribute to the link between temporal impulsivity and obesity in men. *bioRxiv*. 2020.
30. **Morys F***, García-García I*, Dagher A. Nucleus accumbens volume is related to obesity measures in an age-dependent fashion. *Journal of Neuroendocrinology*. 2019.
31. **Morys F**, Janssen LK, Cesnaite E, Beyer F, Garcia-Garcia I, Kube J et al. Hemispheric asymmetries in resting-state EEG and fMRI are related to approach and avoidance behaviour, but not to eating behaviour or BMI. *Human Brain Mapping*. 2019.
32. **Morys F***, Mehl N*, Villringer A, Horstmann A. Unhealthy yet avoidable — how cognitive bias modification alters behavioral and brain responses. *Nutrients*. 2019.
33. Oliva R, **Morys F**, Horstmann A, Castiello U, Begliomini C. The impulsive brain: Neural underpinnings of binge eating behavior in normal-weight adults. *Appetite*. 2019.
34. García-García I, Kube J, **Morys F**, Schrimpf A, Kanaan ASAS, Gaebler M et al. Liking and left amygdala activity during food versus nonfood processing are modulated by emotional context. *Cognitive Affective and Behavioral Neuroscience*. 2019.
35. Oliva R, **Morys F**, Horstmann A, Castiello U, Begliomini C. Characterizing impulsivity and resting-state functional connectivity in normal-weight binge eaters. *International Journal of Eating Disorders*. 2019.
36. **Morys F***, Lammert M*, Hartmann H, Janssen LK, Horstmann A. MIS – A new scoring method for the operation span task that accounts for Math, remembered Items and Sequence. *PsyArXiv*. 2019.
37. **Morys F**, Bode S, Horstmann A. Dorsolateral and medial prefrontal cortex mediate the influence of incidental priming on economic decision making in obesity. *Scientific Reports*. 2018.

* Shared first authorship

ORAL PRESENTATIONS

1. Li Z., Sun Y., Sung J., Bouteldja A., **Morys F.**, Small D. Dietary Restraint and Bingeing Trait's Effects on Food Valuation. Talk presented at the 27th Annual Neuropsychology Day. Montreal, Canada. 11.05.2026

2. Heshmati S., Goldberg J., Stevens-Green S., Faraj M., Tardif C., Gauthier C., Wen Hu X., Dagher A., **Morys F.**, Small D. A MRI Study of Metabolic and Cerebrovascular Health. Talk presented at the annual MeDiC Research Day, Montreal, Canada. 13.11.2025.
3. Sung J., **Morys F.**, Canna A., Davis X., Small D. An influence of BMI and high fat diet on food valuation. Talk presented at the Annual Meeting of the Society for the Study of Ingestive Behavior, Oxford, United Kingdom. 28-07-2025 – 02-08-2025.
4. Kanyamibwa A., **Morys F.**, Fängström D, Tweedale M, Pastor-Bernier A, Azizi H, Liu L, Horstmann A, Dagher A. Ultra-processed food consumption affects structural integrity of feeding-related brain regions independent and via adiposity. Talk presented at the Annual Meeting of the Society for the Study of Ingestive Behavior, Oxford, United Kingdom. 28-07-2025 – 02-08-2025.
5. Quillian J., Attuquayfio T., Sung J., Ko T., Davis X., Maciejewski K., Li F., Santoro N., Kullman S., Preissl H., **Morys F.**, Canna A., Dagher A., Caprio A., Small D. No association between peripheral glucose control and central insulin sensitivity or cognition in youth with overweight/obesity. Talk presented at the Annual Meeting of the Society for the Study of Ingestive Behavior, Chicago, Illinois, USA. 09-07-2024 – 13-07-2024.
6. **Morys F.**, Liu L., Senkevich K., Tremblay C., Rahayel S., Gan-Or Z., Dagher A. (2023). Influence of obesity and MAPT/ApoE genotypes on brain structure and cognition. Talk presented at the Annual Meeting of the Society for the Study of Ingestive Behavior, Portland, Oregon, USA. 11-07-2023 – 15-07-2023.
7. **Morys F.**, Shishikura M., Tremblay C., Hansen J., Misic B., Dagher A. (2022). Neuromolecular underpinnings of human obesity. Talk presented at the Annual Meeting of the Society for the Study of Ingestive Behavior, Porto, Portugal. 12-07-2022 – 17-07-2022.
8. Yu, J., **Morys, F.**, Dagher, A., Kaminska, M. (2021). Associations between sleep -disordered breathing, brain structure and cognitive decline in obesity. Talk presented at the Quebec Respiratory Health Research Network Meeting (virtual). 10-11-2021 – 11-11-2021.
9. **Morys, F.**, Yu, E., Shishikura, M., Vainik, U., Gan-Or, Z., Dagher A. (2021). Cortical thickness, volume, and white matter's fractional anisotropy confer genetic risk for obesity in adolescence. Talk presented at the Annual Meeting of the Society for the Study of Ingestive Behavior (virtual). 12-07-2021 – 16-07-2021.
10. Tremblay C., Rahayel S., Vo A., **Morys F.**, Shafiei G., Markello R.D., Misic B., Dagher A. (2021). Virtual histology of brain atrophy progression in de novo Parkinson's disease. Talk presented at the 15th International Conference on Alzheimer's & Parkinson's Diseases, Virtual Conference.
11. Tremblay C., Rahayel S., Vo A., **Morys F.**, Shafiei G., Markello R., Misic B., Gan-Or Z., Dagher A. (2021). Progression of brain atrophy in Parkinson's disease is shaped by connectome architecture and local vulnerability. Talk presented at the International Congress of Parkinson's Disease and Movement Disorders Virtual Conference (Top abstract).
12. **Morys, F.**, Dadar, M., Dagher, A. (2020). Obesity impairs cognitive function via metabolic syndrome and cerebrovascular disease: an SEM analysis in 15,000 adults from the UK Biobank. Talk presented at a virtual conference: Brain Plasticity - Influences of Lifestyle and Aging, organized by the Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Germany. 12-10-2020 – 13-10-2020.
13. **Morys, F.**, Dagher, A. (2020). The relationship between BMI and volume of subcortical structures is age-dependent. Talk presented at the 2020 Organization for Human Brain Mapping Annual Meeting (virtual). 26-06-2020 - 2020-06-30.
14. Horstmann, A., Mehl, N., **Morys, F.**, & Villringer, A. (2018). Cognitive bias modification alters behavioral and brain responses to food cues in obesity. Talk presented at 51. Congress of the German Psychological Society. Frankfurt a.M., Germany. 2018-09-15 — 2018-09-20.
15. Mehl, N., **Morys, F.**, Villringer, A., & Horstmann, A. (2018). Veränderung automatischer Verhaltenstendenzen bei Adipositas – Was passiert im Gehirn? Talk presented at the 6th Scientific Congress of the German Society for Eating Disorders. Munich, Germany. 2018-03-01 - 2018-03-03.

16. **Morys, F.**, & Horstmann, A. (2017). Dorsolateral and medial prefrontal cortex mediate influence of incidental priming on economic decisions in obesity. Talk presented at 25th Annual Meeting of the Society for the Study of Ingestive Behavior; Montréal, QC, Canada. 2017-07-18 — 2017-07-22.
17. Mehl, N., **Morys, F.**, Villringer, A., Horstmann, A. (2017). 'Neural correlates of retraining automatic action tendencies in obesity'. Talk presented at 25th Annual Meeting of the Society for the Study of Ingestive Behavior; Montréal, QC, Canada. 2017-07-18 — 2017-07-22.

POSTER PRESENTATIONS

1. Heshmati S., Goldberg J., Stevens-Green S., Faraj M., Tardif C., Gauthier C., Wen Hu X., Dagher A., **Morys F.**, Small D. (2026). A MRI study of cerebrovascular and metabolic health. Poster presented at the Canadian Association for Neuroscience annual meeting. Montreal, Canada.
2. Heshmati S., Goldberg J., Stevens-Green S., Faraj M., Tardif C., Gauthier C., Wen Hu X., Dagher A., **Morys F.**, Small D. (2026). Disentangling the Effects of Diet and Adiposity on Brain Structure and Cognition in Midlife: A Multimodal MRI Study. Poster presented at the 11th Canadian Neurometabolic Meeting. Montreal, Canada.
3. Sung J., **Morys F.**, Small D. (2026) Inverse effects of diet and adiposity on nucleus accumbens volume in humans. Poster presented at the Canadian Association for Neuroscience Annual Meeting. Montreal, Canada.
4. Li Z., Sun Y., Sung J., Bouteldja A., **Morys F.**, Small D. (2026) Dietary Restraint and Bingeing Trait's Effects on Food Valuation. Poster presented at the undergraduate research day at McGill University. Montreal, Canada. 11.05.2026
5. Heshmati S., Goldberg J., Stevens-Green S., Faraj M., Tardif C., Gauthier C., Wen Hu X., Dagher A., **Morys F.**, Small D. (2026). A MRI Study of Metabolic and Cerebrovascular Health. Poster presented at the Neuropsychology Day of the Montreal Neurological Institute. Montreal, Canada.
6. Azizi H, Pastor-Bernier A, Tremblay C, Abbasi N, Liu L, Senkevich K, Pourmajidian M, **Morys F**, Vo, Savadjiev P, Yu E, Poline JB, Gan-Or Z, Zeighami Y, Dagher A. (2025). Genetic architecture of Parkinson's disease: investigating the relationships between pathway-specific polygenic risk scores and neuroanatomical features. Poster presented at the Alzheimer's Association International Conference. Toronto, Canada. (https://doi.org/10.1002/alz70856_103914)
7. Heshmati S., Goldberg J., Stevens-Green S., Faraj M., Tardif C., Gauthier C., Wen Hu X., Dagher A., **Morys F.**, Small D. (2026). A MRI Study of Metabolic and Cerebrovascular Health. Poster presented at the Canadian Obesity Summit. Montreal, Canada.
8. Heshmati S., Goldberg J., Stevens-Green S., Faraj M., Tardif C., Gauthier C., Wen Hu X., Dagher A., **Morys F.**, Small D. (2025). A MRI Study of Metabolic and Cerebrovascular Health. Poster presented at the annual MeDiC Research Day. Montreal, Canada.
9. Morys F., Hartmann H, Dagher A, Horstmann A. (2025). Structural brain differences in metabolically healthy and metabolically unhealthy obesity in the UK Biobank sample. Poster presented at the Annual Meeting of the Society for the Study of Ingestive Behavior. Oxford, United Kingdom.
10. Morys F., Tweedale M., Pastor-Bernier A., Azizi H., Tremblay C., Dagher A. (2024). Relationship between obesity, related comorbidities, and subcortical neuroinflammation in young and older adults. Poster presented at the Organization for Human Brain Mapping (OHBM) Annual Meeting. Seoul, South Korea. (<https://doi.org/10.52294/001c.120596>)
11. Tremblay C., Rahayel S., Pastor-Bernier A., St-Onge F., Vo A., Rheault F., Daneault V., **Morys F.**, Rajah N., Villeneuve S., Dagher A. (2024). Atrophy Progression in People With a Family History of Alzheimer's disease is Shaped by Connectivity. Poster presented at the Organization for Human Brain Mapping (OHBM) Annual Meeting. Seoul, South Korea.

12. **Morys F.**, Liu L., Senkevich K., Tremblay C., Rahayel S., Gan-Or Z., Dagher A. (2023). Influence of obesity and MAPT/ApoE genotypes on brain structure and cognition. Poster presented at the Organization for Human Brain Mapping (OHBM) Annual Meeting. Montreal, Canada.
13. Shishikura M., **Morys F.**, Dagher A. (2023). Link between brain structure and genetic predisposition for risky and impulsive behavior in children. Poster presented at the Organization for Human Brain Mapping (OHBM) Annual Meeting. Montreal, Canada.
14. Pastor-Bernier A., Savadjiev P., Poline JB., St-Onge E., Zhang F., O'Donnell LJ., Pasternak O., Gan-Or Z., **Morys F.**, Dagher A. (2023) Influence of body mass index on white matter structure. Poster presented at the Organization for Human Brain Mapping (OHBM) Annual Meeting. Montreal, Canada.
15. **Morys F.**, Shishikura M., Tremblay C., Hansen J., Misic B., Dagher A. (2022). Neuromolecular underpinings of human obesity. Poster presented at the Organization for Human Brain Mapping (OHBM) Annual Meeting. Glasgow, UK
16. Shishikura, M., **Morys, F.**, Yu, E., Dagher, A. (2022). Differentiating subgroups in overweight and obese children using data-driven clustering approach with brain imaging data. Poster presented at the Annual Meeting of the Society for the Study of Ingestive Behavior, Porto, Portugal.
17. Kaminska, M., Yu, J., **Morys, F.**, Ock, E., Lajoie, A., Dagher, A. (2022). Associations between hypersomnolence, obesity, brain health and cognitive decline. Poster presented at the American Thoracic Society meeting.
18. Kouchache, T., Khundrakpam B. S., Vainik, U., Tremblay, C., Kirschner, M., **Morys, F.**, Vo, A., Dagher, A. (2022) Brain regions associated with differences in polygenic risks scores for smoking in typical development. Poster presented at the Organization for Human Brain Mapping (OHBM) Annual Meeting. Glasgow, UK.
19. Kouchache, T., Khundrakpam B. S., Vainik, U., Tremblay, C., Kirschner, M., **Morys, F.**, Vo, A., Dagher, A. (2022) Brain regions associated with differences in polygenic risks scores for smoking in typical development. Poster presented at the Cognitive Neuroscience Society Annual Meeting. San Francisco, US.
20. **Morys F.**, Yu E., Shishikura M., Vainik U., Gan-Or Z., Dagher A. (2021). Cortical Thickness, Volume And White Matter's Fractional Anisotropy Confer Genetic Risk For Obesity In Adolescence. Poster presented at the Annual Meeting of the Society for the Study of Ingestive Behavior (virtual).
21. Hartmann, H., Janssen, L., Herzog, N., **Morys, F.**, Fallon, S., Horstmann, A. (2021). Associations of dietary fat and sugar and bodyweight with dopamine dependent stability and flexibility of working memory representations. Poster presented at the Annual Meeting of the Society for the Study of Ingestive Behavior (virtual).
22. **Morys F.**, Yu E., Shishikura M., Gan-Or Z., Dagher A. (2021). Obesity genetic risk is related to weight gain via structural changes in the frontoparietal network. Poster presented at the 2021 Organization for Human Brain Mapping Annual Meeting (virtual).
23. Tremblay C., Rahayel S., Vo A., **Morys F.**, Shafiei G., Markello R.D., Misic B., Gan-Or Z., Dagher A. (2021). Atrophy Progression in Parkinson's Disease is Associated with Connectivity and Synaptic Activity. Poster presented at the 2021 Organization for Human Brain Mapping Annual Meeting (virtual).
24. Tremblay C., Rahayel S., Vo A., **Morys F.**, Shafiei G., Markello R.D., Dagher A. (2020). Connectome architecture influences brain atrophy progression in de novo Parkinson's disease. Poster presented at the Grand Challenges in Parkinson's Disease Virtual Conference.
25. **Morys, F.**, Dagher A. (2020). The relationship between BMI and volume of subcortical structures is age-dependent. Poster presented at the 2020 Organization for Human Brain Mapping Annual Meeting (virtual).
26. **Morys, F.**, Dadar M., Dagher A. (2020). Obesity impairs cognitive function via metabolic syndrome and cerebrovascular disease: an SEM analysis in 15,000 adults from the UK Biobank. Poster presented at a

virtual conference: Brain Plasticity - Influences of Lifestyle and Aging, organized by the Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Germany.

27. Yau, Y., **Morys, F.**, Dagher, A. (2020). The Dynamics of Brain Connectivity at Rest Underpins Performance in Task. Poster presented at the 2020 Winter Conference on Brain Research, Big Sky, Montana, USA.
28. **Morys, F.**, Janssen, L., Cesnaite, E., Garcia-Garcia, I., Kube, J., Schrimpf, A., et al. (2018). Hemispheric bias in resting state EEG and fMRI is related to approach/avoidance behaviors, but not BMI. Poster presented at 2018 Organization for Human Brain Mapping Annual Meeting; Singapore.
29. **Morys, F.**, & Horstmann, A. (2017). Dorsolateral and medial prefrontal cortex mediate influence of incidental priming on economic decisions in obesity. Poster presented at 33rd Annual Conference of the German Association for the Study of Obesity; Potsdam, Germany.
30. **Morys, F.**, Mehl, N., Horstmann, A., & Villringer, A. (2017). Neural correlates of cognitive bias modification in obesity. Poster presented at 33rd Annual Conference of the German Association for the Study of Obesity; Potsdam, Germany.
31. **Morys, F.**, Simmank, J., & Horstmann, A. (2016). BMI mediates the relationship between temporal impulsivity and cognitive restraint in women. Poster presented at 32nd Annual Conference of the German Association for the Study of Obesity; Frankfurt, Germany.
32. **Morys, F.**, Simmank, J., & Horstmann, A. (2016). BMI mediates the relationship between temporal impulsivity and cognitive restraint in women. Poster presented at 24th Annual Meeting of the Society for the Study of Ingestive Behaviour; Porto, Portugal.
33. **Morys, F.**, Bode, S., Murawski, C., & Horstmann, A. (2015). Influence of incidental priming with visual and gustatory cues on intertemporal choice in obesity. Poster presented at LIMIOR — 1st Leipzig International Meeting for Interdisciplinary Obesity Research; Leipzig, Germany.

AD-HOC REVIEWER

- | | |
|---|---|
| 1. Clinical Nutrition | 18. Scientific Reports |
| 2. Communications Biology | 19. Obesity |
| 3. Brain Structure and Function | 20. Obesity Reviews |
| 4. BMC Psychiatry | 21. Biological Psychology |
| 5. Imaging Neuroscience | 22. Neuroscience and Biobehavioral Reviews |
| 6. Appetite | 23. Wiley Interdisciplinary Reviews: Cognitive Science |
| 7. Journal of Alzheimer's Disease | 24. Current Psychology |
| 8. Aging and Disease | 25. Cerebral Cortex |
| 9. The Lancet Regional Health – Western Pacific | 26. Neurobiology of Stress |
| 10. International Journal of Obesity | 27. International Journal of Environmental Research and Public Health |
| 11. Brain Imaging and Behavior | 28. EBioMedicine |
| 12. Pediatric Obesity | 29. Physiology and Behavior |
| 13. Human Brain Mapping | 30. Wellcome Open Research |
| 14. Plos One | 31. NPJ Aging |
| 15. Neuroimage | 32. NPJ Parkinson's disease |
| 16. Neurobiology of Aging | |
| 17. Psychosomatic Medicine | |

TECHNICAL SKILLS

Programming languages

- MATLAB, R, Python, SQL, Bash

Neuroimaging tools

- FSL, SPM, Nilearn, Nipype, Nibabel, AFNI, Tractoflow, fMRIPrep, FreeSurfer, CAT

Chosen statistical tools

- General linear model, generalized linear model, machine learning: LASSO regression, partial least squares analysis, principal component analysis, independent component analysis

LANGUAGE PROFICIENCY

1. Polish – native (C2)
2. English – fluent (C2)
3. German – fluent (C1)
4. French – fluent (C1)